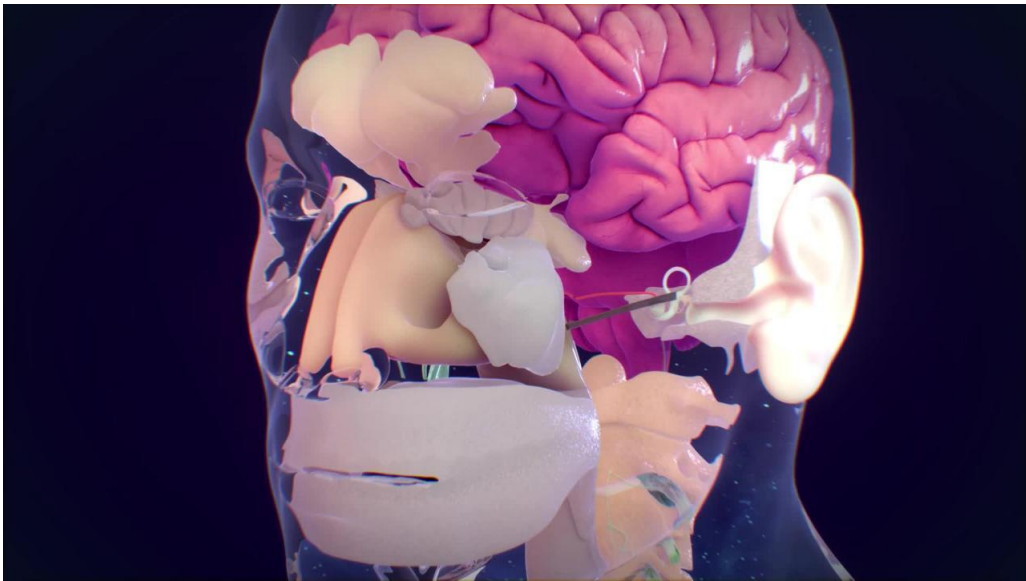
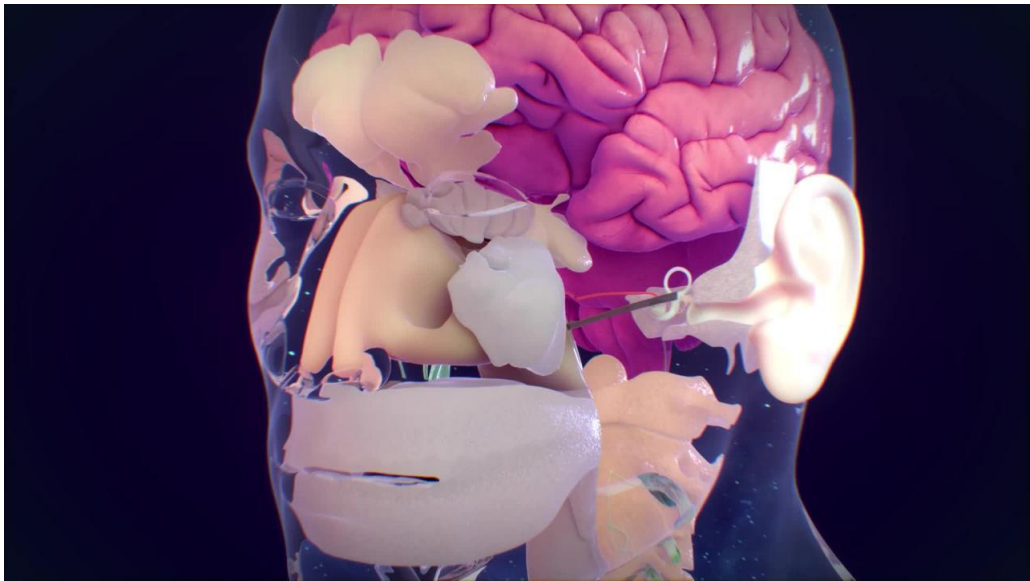










# Loss of Smell

(Anosmia)

By **Marvin P. Fried**, MD, Montefiore Medical Center, The University Hospital of Albert Einstein College of Medicine

Reviewed/Revised Mar 2025 | Modified Aug 2025

Anosmia is complete loss of smell. Hyposmia is partial loss of smell. Most people with anosmia can taste salty, sweet, sour, and bitter substances but cannot tell the difference between specific flavors. The ability to tell the difference between flavors actually depends on smell, not the taste receptors on the tongue. Therefore, people with anosmia often complain of losing their sense of taste and of not enjoying food.

A loss of smell receptors due to aging causes a decreased ability to smell in older adults. People typically notice changes in smell by age 60. After age 70, changes are substantial.

## Causes of Anosmia

Anosmia that is not the result of aging occurs when swelling or another blockage of the nasal passages prevents odors from reaching the olfactory area or when parts of the olfactory area or its connections to the brain are destroyed (see table [Some Causes and Features of Anosmia](#)). The olfactory area, where odors are detected, is located high in the nose (see [How People Sense Flavors](#)).

### Common causes

The most common causes include

- [Head injury](#) (young adults)
- Viral infections
- [Alzheimer disease](#) (older adults)

## Less common causes

Medications can contribute to anosmia in susceptible people. Polyps, tumors, other infections in the nose, and seasonal allergies ([allergic rhinitis](#)) may interfere with the ability to smell. Occasionally, serious infections of the nasal sinuses or radiation therapy for cancer causes a loss of smell or taste that lasts for months or even becomes permanent. These conditions can damage or destroy smell receptors. The role of tobacco is uncertain.

A very few people are born without a sense of smell.

Anosmia or hyposmia may be an early symptom of [COVID-19](#), an acute respiratory illness that can be severe. COVID-19 is caused by the coronavirus called SARS-CoV2.

## Evaluation of Anosmia

The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

### Warning signs

The following findings are of particular concern:

- Recent head injury
- Symptoms of nervous system dysfunction, such as weakness, trouble with balance, or difficulty seeing, speaking, or swallowing

- Sudden start of symptoms
- Known exposure or lack of vaccination to COVID-19

### When to see a doctor

People who have warning signs should see a doctor right away. Other people should see a doctor when possible.

### What the doctor does

Doctors first ask questions about the person's symptoms and medical history and then do a physical examination. What doctors find during the history and physical examination often suggests a cause and the tests that may need to be done (see table [Some Causes and Features of Anosmia](#)).

A common cause of permanent loss of smell is a head injury, as may occur in a car accident. Head injury can damage or destroy fibers of the olfactory nerves (the pair of cranial nerves that connect smell receptors to the brain) where they pass through the roof of the nasal cavity. Sometimes the injury involves a fracture of the bone (cribriform plate) that separates the brain from the nasal cavity. Damage to the olfactory nerves can also result from infections (such as abscesses) or tumors near the cribriform plate.

Another common cause is an upper respiratory infection, especially [influenza](#) (flu). Flu may be the cause in up to one quarter of people with hyposmia or anosmia. [Alzheimer disease](#) and some other degenerative brain disorders (such as [multiple sclerosis](#)) can damage the olfactory nerves, commonly causing loss of smell.

### Less common causes

Medications can contribute to anosmia in susceptible people. Polyps, tumors, other infections in the nose, and seasonal allergies ([allergic rhinitis](#)) may interfere with the ability to smell. Occasionally, serious infections of the nasal sinuses or radiation therapy for cancer causes a loss of smell or taste that lasts for months or even becomes permanent. These conditions can damage or destroy smell receptors. The role of tobacco is uncertain.

A very few people are born without a sense of smell.

Anosmia or hyposmia may be an early symptom of [COVID-19](#), an acute respiratory illness that can be severe. COVID-19 is caused by the coronavirus called SARS-CoV2.

## Evaluation of Anosmia

The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

### Warning signs

The following findings are of particular concern:

- Recent head injury
- Symptoms of nervous system dysfunction, such as weakness, trouble with balance, or difficulty seeing, speaking, or swallowing
- Sudden start of symptoms
- Known exposure or lack of vaccination to COVID-19

## When to see a doctor

People who have warning signs should see a doctor right away. Other people should see a doctor when possible.

## What the doctor does

Doctors first ask questions about the person's symptoms and medical history and then do a physical examination. What doctors find during the history and physical examination often suggests a cause and the tests that may need to be done (see table [Some Causes and Features of Anosmia](#)).

Doctors ask when and how anosmia started and how long it has lasted. They also ask whether it started before or shortly after a cold, bout of flu, or head injury. They note other symptoms such as a runny or stuffy nose and whether any nasal discharge is watery, bloody, thick, or foul-smelling. Doctors check for neurologic symptoms, especially those that involve a change in mental status (for example, difficulty with short-term memory) or the cranial nerves (for example, double vision or difficulty speaking or swallowing).

Questions about the person's medical history involve sinus disorders, head injury or surgery, allergies, medications and illicit drugs used, and exposure to chemicals or fumes.

During the physical examination, doctors inspect the nasal passages for swelling, inflammation, discharge, and polyps. Doctors also do a complete [neurologic examination](#) that is particularly focused on mental status and the cranial nerves.



## Some Causes and Features of Anosmia

| Cause                             | Common Features*                                                                                                                                                                                                    | Diagnostic Approach                                                        |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Blockage within the nose          |                                                                                                                                                                                                                     |                                                                            |
| <a href="#">Nasal allergies</a>   | <p>In people who have chronic allergy symptoms (such as nasal congestion and a clear discharge)</p> <p>No pain</p> <p>Symptoms that often occur during certain seasons or after exposure to specific substances</p> | <p>A doctor's examination</p> <p>Allergy testing (skin or blood tests)</p> |
| <a href="#">Nasal polyps</a>      | <p>Polyps that are usually seen during the examination</p>                                                                                                                                                          | <p>A doctor's examination alone</p>                                        |
| Destruction of smell receptors†   |                                                                                                                                                                                                                     |                                                                            |
| <a href="#">Atrophic rhinitis</a> | <p>Thin, hard, dry mucous membranes</p> <p>Widened nasal passages</p> <p>Crusting inside the nose</p> <p>A foul-smelling odor in the nose</p>                                                                       | <p>Sometimes a doctor's examination alone</p> <p>Sometimes a biopsy</p>    |
| <a href="#">Chronic sinusitis</a> | <p>A thick, foul-smelling nasal discharge most or all of the time</p> <p>Previous sinus infections</p>                                                                                                              | <p>Sometimes a doctor's examination alone</p> <p>Usually CT</p>            |
| <a href="#">COVID-19</a>          |                                                                                                                                                                                                                     |                                                                            |



**MSD**

Copyright © 2026 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. All rights reserved.